HW 2 due: 10/18

	$P(B_i)$	$P(R \mid B_i)$
① B_1	40% sales	0.10
B_2	30%	0.05
B_3	20%	0.03
B_4	10%	0.02

Want to know: $P(B_i \mid R)$.

$$P(R) = P(R|B_1)P(B_1) + P(R|B_2)P(B_2) + P(R|B_3)P(B_3) + P(R|B_4)P(B_4)$$

$$= (0.10)(0.40) + (0.05)(0.30) + (0.03)(0.20) + (0.02)(0.10)$$
$$= 0.04 + 0.015 + 0.006 + 0.002$$
$$= 0.063$$

$$P(B_1|R) = \frac{P(R|B_1)P(B_1)}{P(R)} = \frac{(0.10)(0.40)}{0.063} = \frac{0.04}{0.063} = 0.6349$$

$$P(B_2|R) = \frac{P(R|B_2)P(B_2)}{P(R)} = \frac{(0.05)(0.30)}{0.063} = \frac{0.015}{0.063} = 0.2381$$

$$P(B_3|R) = \frac{P(R|B_3)P(B_3)}{P(R)} = \frac{(0.03)(0.20)}{0.063} = \frac{0.006}{0.063} = 0.0952$$

$$P(B_4|R) = \frac{P(R|B_4)P(B_4)}{P(R)} = \frac{(0.02)(0.10)}{0.063} = \frac{0.002}{0.063} = 0.0317$$

② $P(H) = 0.16$ $P(H|I) = 0.35$
$P(L) = 0.19$ $P(I|N) = 0.11$
$P(I) = 0.17$
$P(N) = 0.65$ (1 − (0.16 + 0.19) normal or regular workweek)
$P(R) = 0.83$

① Find $P(R \cap L) =$

② $P(I|H) = \frac{P(H|I)P(I)}{P(H)} = \frac{(0.35)(0.17)}{0.16}$

$P(I|H) = 0.37$

③ $P(I) = P(I|H)P(H) + P(I|L)P(L) + P(I|N)P(N)$

$0.17 = (0.37)(0.16) + P(I|L)0.19 + (0.11)(0.65)$

$P(I|L) = 0.2068$

$1 - P(I|L) = 1 - 0.2068 = 0.7932$

↑

(4) $P(R \cap L) = P(R|L)P(L)$
$= 0.7932 \cdot 0.19$
$= 0.1507$

15.07% have a regular heartbeat and low BP

(6) 1) $h(z) = \begin{cases} \frac{1}{2}, & \text{if } z = 1 \\ \frac{1}{3}, & \text{if } z = 8 \\ \frac{1}{6}, & \text{if } z = 27 \end{cases}$

1	$X^3 = z = 1$	$f(1) = \frac{3}{6} = \frac{1}{2}$
2	$X^3 = z = 8$	$f(2) = \frac{2}{6} = \frac{1}{3}$
3	$X^3 = z = 27$	$f(3) = \frac{1}{6}$

2) $E[z] = \sum_{z=1,8,27} z \cdot h(z) = (1)(\frac{1}{2}) + (8)(\frac{1}{3}) + (27)(\frac{1}{6}) = \frac{1}{2} + \frac{8}{3} + \frac{9}{2} = \frac{3}{6} + \frac{16}{6} + \frac{27}{6} = \frac{46}{6} = \frac{23}{3}$

(3) $f(2) = P(X=2) = \frac{1}{36}$ $f(3) = P(X=3) = \frac{2}{36} = \frac{1}{18}$ $f(4) = P(X=4) = \frac{3}{36} = \frac{1}{12}$
$f(5) = P(X=5) = \frac{4}{36} = \frac{1}{9}$ $f(6) = P(X=6) = \frac{5}{36}$ $f(7) = P(X=7) = \frac{6}{36} = \frac{1}{6}$
$f(8) = P(X=8) = \frac{5}{36}$ $f(9) = P(X=9) = \frac{4}{36} = \frac{1}{9}$ $f(10) = P(X=10) = \frac{3}{36} = \frac{1}{12}$
$f(11) = P(X=11) = \frac{2}{36} = \frac{1}{18}$ $f(12) = P(X=12) = \frac{1}{36}$ ★ cont'd

(4) 1) $\frac{1}{c} + \frac{2}{c} + \frac{3}{c} + \frac{4}{c} = 1$
$1 + 2 + 3 + 4 = c$
$c = 10$

2) $c \sum_{x=1}^{10} x = 1 + 2 + \cdots n$ $\left\{ \frac{n(n+1)}{2} = \frac{10(11)}{2} = 55 \right.$
$55c = 1$
$c = 1/55$

3) $\sum_{x=1}^{\infty} c(\frac{1}{4})^x = 1 = c \sum_{x=1}^{\infty} (\frac{1}{4})^x = \frac{1}{4} + \frac{1}{4}^2 + \frac{1}{4}^3 \cdots$ (b/c starts at 1) $\left\{ \frac{\frac{1}{4}}{1 - \frac{1}{4}} \right.$ $c(\frac{\frac{1}{4}}{\frac{3}{4}}) = 1$
$c(1/3) = 1$
$c = 3$

4) $c(0+1)^2 + c(1+1)^2 + c(2+1)^2 + c(3+1)^2 = 1$
$c + 4c + 9c + 16c = 1$
$30c = 1$
$c = 1/30$

(5) $f(-1) = \frac{4}{9}$ $f(0) = \frac{1}{9}$ $f(1) = \frac{4}{9}$
$E[X] = -1(\frac{4}{9}) + 0(\frac{1}{9}) + 1(\frac{4}{9}) = -\frac{4}{9} + \frac{4}{9} = 0$
$E[X^2] = (-1)^2(\frac{4}{9}) + (0)^2(\frac{1}{9}) + (1)^2(\frac{4}{9}) = \frac{4}{9} + \frac{4}{9} = \frac{8}{9}$
$E[3X^2 - 2X + 4] = 3E[X^2] - 2E[X] + 4 = 3(\frac{8}{9}) - 2(0) + 4 = \frac{8}{3} + 4 = \frac{20}{3}$

③ cont.

$$f(x) = \begin{cases} \frac{1}{36} & \text{for } x = 2, 12 \\ \frac{1}{18} & \text{for } x = 3, 11 \\ \frac{1}{12} & \text{for } x = 4, 10 \\ \frac{1}{9} & \text{for } x = 5, 9 \\ \frac{5}{36} & \text{for } x = 6, 8 \\ \frac{1}{6} & \text{for } x = 7 \end{cases}$$